MUSCULOSKELETAL INFECTIONS-PART B
MUSCLE INFECTIONS

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LEARNING AND TESTABLE OBJECTIVES

- List the major pathogens of myositis
  - *Streptococcus pyogenes*
  - *Clostridium botulinum*
  - *Clostridium tetani*
  - *Trichinella spiralis*
  - *Cysticercus (Taenia spp.)*

- Discuss pertinent characteristics of the pathogens including,
  - Morphological and cultural characteristics
  - Source, transmission, and relevant epidemiology
  - Virulence factors and pathogenesis
  - Disease association and complications
  - Laboratory Diagnosis
  - Treatment, Prevention

2

INFECTIOUS MYOSITIS
BUG LIST

BACTERIA	FUNGI	PARASITES
Staphylococcus aureus	*Candida spp*	*Trichinella spiralis*
Streptococcus pyogenes	*Others:* *Cryptococcus spp*	Cysticerci of *Taenia spp* *(tapeworm larvae)*
Clostridium botulinum		*Others: Trypanosoma cruzi* (causes Chaga's disease); *Toxoplasma gondii*
Clostridium tetani		
Others- Borrelia burgdorferi *Mycobacteria, other non- sporing anaerobes*		

BACTERIAL INFECTIONS

Pyomyositis	Necrotizing fasciitis	Necrotizing myositis	Tetanus
• Purulent condition of the skeletal muscles. • Localized to single muscle group. • Polymicrobial/ Monomicrobial • #1 pathogen: *Staphylococcus aureus*	• Rare but rapidly progressive and fatal. • Involves extremities, abdominal wall or perineum • Poly/ Mono microbial • Pathogen: *Streptococcus pyogenes*(GAS- Group A Strep) causes majority cases.	• GAS GANGRENE or clostridial myonecrosis • Uncommon but rapidly progressive and life threatening • Polymicrobial • Several species of Clostridia: *C. perfringens, C.septicum, C.novyi*, etc. along with GPC and GNB.	• Neuromuscular condition • **Pathogen-** ***Clostridium tetani*** • Spastic paralysis due to toxins • can be fatal

NECROSTIZING FASCITIS

- A rare bacterial infection that spreads quickly.
- Associated with cuts/trauma/burns/ patients with diabetes
- Poly/ Monomicrobial. Most important pathogens:
  - Group A Streptococci (GAS)- *S.pyogenes (referred to as flesh eating bacteria)*
  - Can also involve MRSA, Enterococci, GNB etc.

STREPTOCOCCUS PYOGENES (GROUP A STREPTOCOCCI AKA GAS) MORPHOLOGY AND CULTURE

- Gram positive cocci in chains

Beta-hemolytic on blood agar

Catalase negative

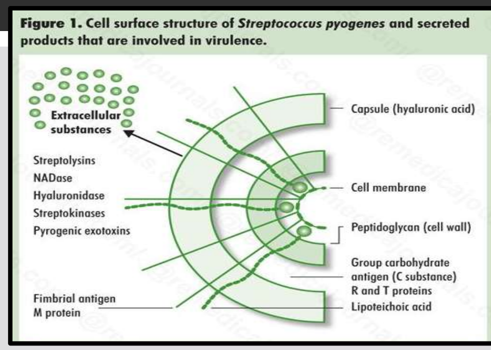
Sensitive to bacitracin

The Big Picture

Diseases caused by *Streptoccus pyogenes*

- Skin and soft tissue infections- cellulitis, Necrotizing fasciitis, erysipelas
- Strep throat
- Toxic Shock Syndrome
- Non suppurative sequelae- RHD and AGN

VIRULENCE FACTORS PATHOGENESIS



Entry of Strep via wound	Adherence to epithelial cells at site of wound LTA-M protein complex bind to Fibronectin	Multiplication while evading phagocytosis M proteins esp. M1 and M3 types	Rapid spread due to hyaluronidase and hydrolytic enzymes Further damage due to toxins like Streptolysins

Possible Complication-Toxic Shock Syndrome
(Vir Fac- Streptococcal Pyrogenic Exotoxin (TSST)

CLINICAL FEATURES
DIAGNOSIS, TREATMENT

C/F

- Fulminant tissue destruction, systemic signs of toxicity, and high mortality.
- Fever, extensive and rapidly progressive redness, pain, swelling, bullae and necrosis of area involved-usually limbs.
- Complications- TSS/sepsis and organ failure
- Important differential diagnosis for Gas gangrene

Diagnosis and Treatment

- Tissue biopsy culture, Blood culture-positive
- Blood tests-Leukocytosis, CRP elevated, Serum Creatine Kinase and AST elevated
- Imaging and blood work up
- Treatment: Antibiotics (Clindamycin+ Vancomycin+ Carbapenem)broad spectrum to cover Strep, Staph, GNB and Anaerobes) + Surgical debridement

Clostridium perfringens and *Clostridium tetani*

Clostridia

- *C. perfringens*
- *C. tetani*
- *C. botulinum*
- *C. difficile*

Spore-Forming, Anaerobic, Gram-Positive Bacilli

9

GAS GANGRENE/ CLOSTRIDIAL MYONECROSIS

Lawn mover
injury to left foot

Right arm with open wound-
Combat related injury

High risk settings for Gas Gangrene

Posttraumatic exposure to soil
- Open wounds or fractures
- Gunshot wounds with foreign bodies
- Burns and frostbites
- IV drug abuse

Postoperative
- Bowel resection
- Biliary surgery

Spontaneous
- Colon Cancer(*C. septicum*)
- Neutropenia

PATHOGENESIS OF GAS GANGRENE

Contamination of deep wounds → Germination of bacterial spores → Conversion to vegetative forms

- fermentation of carbohydrates in tissues
  - Abundant gas formation
- production of necrotizing and hemolytic exotoxins
  - 12 toxins. Most imp. Alpha toxin also called Lecithinase or Phospholipase C
- production of extracellular enzymes:
  - Collagenase and Hyaluronidase

1-3 days after injury
- ✓ Acute and increasing pain
- ✓ Edema, gas (crepitus in tissue)
- ✓ Discoloration and hemorrhagic bullae
- ✓ Systemic signs include fever, tachycardia, hypotension, shock
- ✓ Rapidly progressive and can be fatal

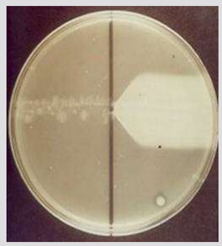
Alpha toxin leads to lysis of RBCs, WBCs, muscle cells, etc. There is increased vascular permeability, systemic absorption of toxins ➡ hypotension ➔ shock

GAS GANGRENE- DIAGNOSIS

- Clinical diagnosis- discoloration, bullae, dishwater pus discharge, crepitus
- Radiograph-linear streaks of gas in soft tissue/ bubbles in tissue
- Grams stain-Wound swab shows GPB with box car morphology. Paucity of neutrophils, spores absent.
- Anaerobic culture- target beta hemolysis or double zone beta hemolysis.
- Identification tests

Blood agar double zone of hemolysis

With anti-α toxin



No anti-α toxin

Nagler reaction to demonstrate *lecithinase* *(see notes section for details)*

Stormy fermentation in milk medium

Nagler's reaction or test is used to detect the alpha toxin responsible for gas gangrene. In Nagler's test, one half of the Egg Yolk agar is plate coated with antitoxin while the other half isn't. On that half of the plate where there is no antitoxin, opaque areas are seen around the colonies due to the action of toxin lecithinase (Phospholipase C) elaborated by *Cl.perfringens* on the lecithin present in Egg Yolk medium.

A CASE OF TETANUS

- A 12-year-old boy presents with stiffness of the jaw and neck along with inability to swallow.
- Twelve days ago he stepped on a rusty nail. This led to a punctured wound and the area is now swollen, red and painful. He has been experiencing tingling numbness and spasms in his calf muscles.
- O/E he has trismus (jaw rigidity), facial muscle spasms, dysphagia and neck rigidity. His back is arched producing opisthotonus.
- Labs: CBC-leukocytosis, CSF, Blood chemistries-normal
- Diagnosis: Tetanus (generalized)

- What is tetanus?
- What is the causal organism and its features?
- What is the virulence factor and its mode of action?
- How is the disease diagnosed, treated and prevented?

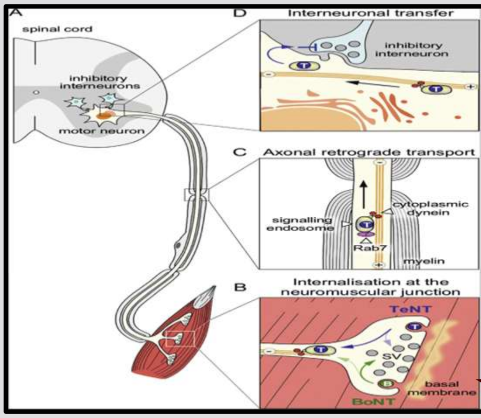
C.tetani- CHARACTERISTICS

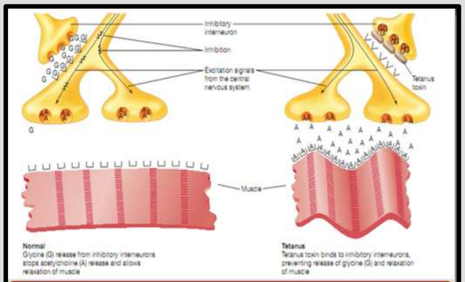
Gram stain of culture smear of *C.tetani. Note the GPB with terminal spores giving them a tennis racket/ drum-stick appearance.*

- GBP with terminal spores (tennis racket/ drum-stick appearance)
- Obligate anaerobe.
- Found in animal gut and contaminates soil/ dust (spores viable for many years)
- Transmission- puncture wounds/ trauma, needs low tissue oxygenation
- Elaborates important Toxins:
  - Tetanospasmin/Tetanus Neurotoxin(TeNT)-an AB exotoxin, high potency with lethal dose 2.5ng/kg.
  - Tetanolysin

TETANUS-PATHOGENESIS

Anaerobic conditions ➔ spore germination ➔ local bacterial multiplication ➔ toxin production ➔ toxin enters nervous system peripherally and is carried via retrograde axonal transport to CNS





- Irreversibly binds receptors and cleaves SNARE proteins ➔ blocks release of glycine and GABA (inhibitory neurotransmitters) ➔ unopposed firing of the motor neurons ➔ constant contraction

Mode of action of Tetanus Toxin

TETANUS- CLINICAL FEATURES, DIAGNOSIS AND PREVENTION

Wound
- **No or minimal inflammation at the site of infection**

Clinical findings
- **Spastic paralysis = strong muscle spasms**
- **Early symptoms**
  - **Lockjaw (trismus) = spasms of masseter muscles**
  - **Risus sardonicus = grimace**
  - **Difficulty swallowing = pharyngeal muscle spasms**
- **Late findings**
  - **Muscle rigidity**
  - **Opisthotonus = arching of back**
- **Others- Generalized tetanus, Neonatal tetanus**

Risus sardonicus

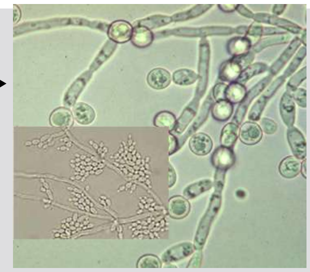
Lock jaw

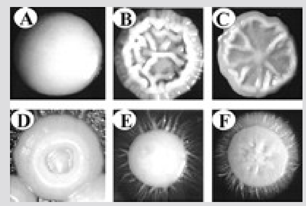
Opisthotonus

Treatment- wound treatment, antibiotics, muscle relaxants
Prevention- passive(tetanus immunoglobulins) and active immunization (Tetanus Toxoid).

FUNGAL MYOSITIS PATHOGENS

- *Candida* species
  - Most common cause
  - Budding yeast with blastoconidia
  - Smooth, white, creamy colonies
  - May undergo phenotype switching to fuzzy colonies and hyphae/pseudohyphae



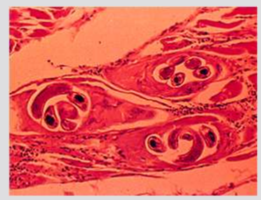


- Increased risk due to
  - ↓ granulocytes
  - Chemotherapy
  - ↑ broad-spectrum antibiotics
- Often seen in conjunction with chronic mucocutaneous candidiasis
- Treatment = amphotericin B or azole

PARASITIC MYOSITIS
Trichinellosis/ Trichinosis and Cysticercosis

Trichinella spiralis – PORK WORM

- Nematode parasite which cause Trichinellosis/ Trichinosis.
- In the US, ~1.5 million persons carry live *Trichinella* in musculature
- Risk factors: Consumption of raw or undercooked meat which is usually not sourced from commercially farmed food animals.
- Typical sources: undercooked pork/ wild game meat-bear, wild boar,wildcat,fox,wolf, seal or walrus.
- Homemade jerkies or sausages from such meat.
- Infective form- Larvae encysted within muscles



T. spiralis encysted in muscle

22

Morbidity and Mortality Weekly Report (*MMWR*)

Search

Outbreak of Human Trichinellosis — Arizona, Minnesota, and South Dakota, 2022

Weekly / May 23, 2024 / 73(20);456–459

Print

Shama Cash-Goldwasser, MD[1]; Dustin Ortbahn, MPH[2]; Muthu Narayan, DO[3]; Conor Fitzgerald, MPH[4]; Keila Maldonado[5]; James Currie, MD[6]; Anne Straily, DVM[7]; Sarah Sapp, PhD[7]; Henry S. Bishop[7]; Billy Watson, PhD[7]; Margaret Neja[7]; Yvonne Qvarnstrom, PhD[7]; David M. Berman, DO[8]; Sarah Y. Park, MD[8]; Kirk Smith, DVM, PhD[9]; Stacy Holzbauer, DVM[9,10] (VIEW AUTHOR AFFILIATIONS)

View suggested citation

Summary

What is already known about this topic?

Human trichinellosis cases in the United States are rare and are usually acquired through consumption of wild game.

What is added by this report?

Among eight persons who shared a meal that included the meat of a black bear harvested in Canada and frozen for 45 days, six trichinellosis cases were identified. The meat was grilled with vegetables and served rare; two cases occurred in persons who ate only the vegetables. Motile freeze-resistant *Trichinella nativa* larvae were identified in remaining meat frozen for >15 weeks.

What are the implications for public health practice?

Cooking meat to an internal temperature of ≥165°F (≥74°C) is necessary to kill *Trichinella* spp. parasites. *Trichinella*-infected meat can cross-contaminate other foods, and raw meat should be kept and prepared separate from other foods to prevent cross-contamination.

Figure

FYI

Morbidity and Mortality Weekly Report

Two Outbreaks of Trichinellosis Linked to Consumption of Walrus Meat — Alaska, 2016–2017

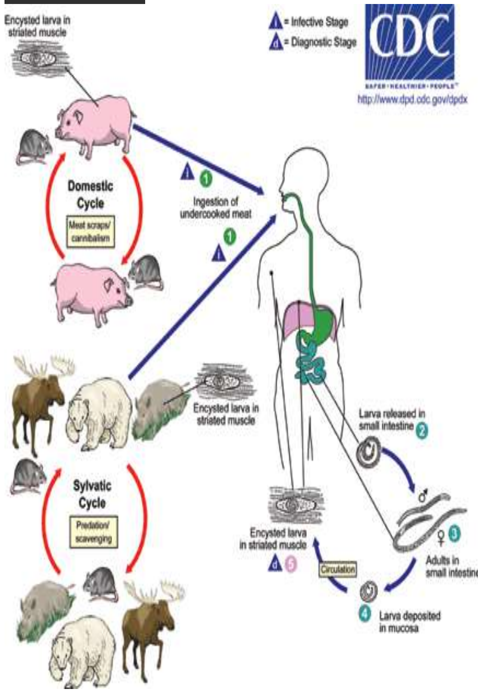
Yuri P. Springer, PhD[1,2]; Shannon Casillas, MPH[3]; Kathryn Helfrich, MSN, MPH[1]; Deanna Mocan[4]; Marscleite Smith[4]; Gabriela Arriaga[4]; Lyndsey Mixson[1,5]; Louisa Castrodale, DVM[1]; Joseph McLaughlin, MD[1]

During 1975–2012, CDC surveillance identified 1,680 trichinellosis cases in the United States with implicated food items; among these cases, 1,219 were attributed to consumption of raw or pork products, and 461 were attributed to nonpork products.

On September 19, staff members of Norton Sound Regional Hospital in Nome reported two additional suspected trichinellosis cases from the same community, in the adult aunt (patient D) and uncle (patient E) of patient A. Both patients

Reference: MMWR / July 7, 2017 / Vol. 66 / No. 26

Life cycle and Pathogenesis



MOI: Ingestion of meat containing larvae

Mild to severe infection based on load of larvae and immune status of patient.

1-2 days- Enteral phase.
Release of larvae from meat>penetrate intestinal mucosa and mature into adult worms > copulation and release of larvae.

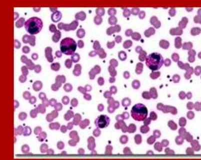
2-8 days later- Parenteral phase
Larvae migrate via blood stream to skeletal muscles.

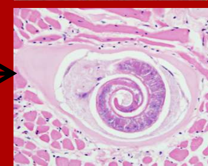
Enteral phase: nausea, abdominal pain, diarrhea.
Parenteral phase:

- Muscle pain, swelling, weakness.
- Subungual splinter hemorrhages.
- Eosinophilia
- More severe cases- myocarditis, encephalitis, thrombo-embolic phenomenon.
- Immune response leads to adult worm expulsion, but larvae can persist for months-years.

TRICHINOSIS

- **Diagnosis**
  - Recent history of eating raw or undercooked meat
  - Clinical presentation- including eosinophilia
  - Detection of specific antibodies in the serum
  - Muscle biopsy(infrequently done) for histopath microscopy to visualize encysted larvae. Also helps in molecular detection of species.





- **Treatment**
  - Corticosteroid for treatment of inflammatory symptoms in severe disease
  - Albendazole or Mebendazole
- **Prevention**
  - Adequate cooking of meat

26

TAENIA SOLIUM – PORK TAPEWORM

- *T. solium* is a cestode- Tape worm with a scolex (head), flat body, segmented proglottids.
- Adult tape worm infestation is Taeniasis (acquired by eating measly pork)

Scolex and body of adult tapeworm

- Skin and soft tissue manifestations are seen due to Cysticercosis caused by the larvae of tapeworms (and not by the adult worms).

- Mode of infection: Ingestion of tapeworm eggs shed by another human being suffering from Taeniasis.

- Eggs hatch in stomach and release oncosphere. These penetrate intestinal wall ➡ circulation ➡ travel to tissue(muscles, CNS, eyes) and develop into cysticerci.

Note the cysticerci over the trunk and shoulders

CYSTICERCOSIS

- Clinical picture
  - Cysticerci in muscle: asymptomatic or nodules, inflammation
  - Cysticerci in the eyes: loss of visual acuity
  - Neurocysticercosis: seizures, and headaches

- Diagnosis
  - Radiology
  - Muscle biopsy- histopathology
  - Antibody detection

- Treatment
  - Albendazole or praziquantel
  - Steroids to decrease inflammation
  - Surgical removal of cysts

Muscle biopsy- histopathology section. Microscopy shows encysted larvae

28

REFERENCES

- Murray's Medical Microbiology 9th edition
  - *Staphylococcus aureus* – Ch.18
  - *Clostridium* spp – Ch.30
  - *Neisseria gonorrhoeae* – Ch.23
  - *Borrelia burgdorferi* – Ch.32
  - *Candida* spp. – Ch.65
  - *Trichinella spiralis* – Ch.75
  - *Taenia solium* – Ch.77
- CDC website for parasites
- www.uptodate.com


THANK YOU